

Equation Sheet: Electrostatics

Intro Physics II • Lectures 6–13 • Knight ch. 22–25

fundamental = know cold (your starting points); **derived** = a line or two from the fundamentals.

Symbols and constants

q, Q charge (C)	Φ electric flux (N·m ² /C)	U potential energy (J)
$\vec{E}$ electric field (N/C = V/m)	λ, η, ρ line / area / volume charge density	V potential (V = J/C)
r, d distance / separation (m)	A area	k 8.99×10^9 N·m ² /C ²
ϵ_0 8.85×10^{-12} C ² /(N·m ²),	$k = 1/(4\pi\epsilon_0)$	$e = 1.60 \times 10^{-19}$ C

1. Coulomb's law & the electric field (Knight 22.4–22.5, 23.1–23.2)

fundamental

Force between two point charges
(along the line joining them; repulsive for like signs; $F \propto 1/r^2$)

$$F = \frac{k |q_1 q_2|}{r^2}, \quad k = \frac{1}{4\pi\epsilon_0}$$

Field — force per unit test charge

$$\vec{E} = \frac{\vec{F}_{\text{on } q}}{q}, \quad \vec{F} = q\vec{E}$$

Field of a point charge
(away from +, toward −)

$$E = \frac{k |q|}{r^2}$$

Superposition (both are vector sums)

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i, \quad \vec{E}_{\text{net}} = \sum_i \vec{E}_i$$

2. Continuous charge distributions (Knight 23.3–23.6)

fundamental

Chop into pieces, add up (integrate)

$$\vec{E} = \int d\vec{E}, \quad dE = \frac{k dq}{r^2}$$

Charge densities

$$\lambda = \frac{Q}{L}, \quad \eta = \frac{Q}{A}, \quad \rho = \frac{Q}{V}$$

derived

Infinite line of charge (distance r)

$$E = \frac{\lambda}{2\pi\epsilon_0 r} = \frac{2k\lambda}{r}$$

Infinite plane / sheet of charge

$$E = \frac{\eta}{2\epsilon_0} \quad (\text{uniform — no } r)$$

3. Electric flux & Gauss's law (Knight 24.1–24.3)

fundamental

Electric flux through a surface

$$\Phi = \oint \vec{E} \cdot d\vec{A}$$

Gauss's law
(any closed surface; only the *enclosed* charge matters)

$$\Phi = \frac{Q_{\text{enc}}}{\epsilon_0}$$

derived

Uniform field through a flat area

$$\Phi = EA \cos \theta \quad (\theta \text{ from the normal})$$

Charge *outside* a closed surface

$$\Phi_{\text{net}} = 0$$

Sphere / spherical shell, total charge Q
($r > R$)

$$E = \frac{kQ}{r^2} \quad (\text{same as a point charge})$$

4. Conductors in equilibrium (Knight 24.4–24.6)

fundamental

Field inside the conducting material	$\vec{E} = 0$
Excess charge	all on the outer surface
Field at the outer surface	$\perp$ to the surface, $E = \frac{\eta}{\epsilon_0}$

derived

Empty cavity inside a conductor	$\vec{E} = 0$ in the cavity (Faraday shielding)
Charge $+q$ placed in the cavity	$-q$ on the inner wall, $+q$ on the outer surface

5. Electric potential energy & potential (Knight 25.1–25.7)

fundamental

Two point charges (keep the signs; $U \rightarrow 0$ as $r \rightarrow \infty$; one power of r)	$U = \frac{k q_1 q_2}{r}$
Energy conservation	$K_i + U_i = K_f + U_f, \quad W_{\text{elec}} = -\Delta U$
Potential — energy per unit charge	$V = \frac{U}{q}, \quad U = qV \quad (1 \text{ V} = 1 \text{ J/C})$
Point charge (scalar; keep the sign; $V \rightarrow 0$ as $r \rightarrow \infty$)	$V = \frac{k q}{r}, \quad V_{\text{net}} = \sum_i \frac{k q_i}{r_i}$

derived

Potential difference from the field	$\Delta V = - \int_i^f \vec{E} \cdot d\vec{s} \xrightarrow{\text{uniform}} \Delta V = -Ed$
Field from the potential (the gradient)	$E_x = -\frac{dV}{dx}, \quad \vec{E} = -\nabla V$
Equipotentials	$\perp$ field lines; $\vec{E}$ points toward decreasing V

Companion simulations at hoppese.github.io/Intro-physics-sims/: `electroscope`, `e-fields-conductors-and-insulators`, `electrostatics-explorer`, `field-superposition-explorer`, `discrete-charge-builder-3d`, `gauss-flux`, `pe-vs-separation`, `equipotential-explorer`.